Project Report

Rising Waters: A Machine Learning Approach to Flood Prediction

1. INTRODUCTION

1.1 Project Overview

The Flood Prediction System is a machine learning-based web application developed to predict the possibility of severe floods using meteorological parameters such as Cloud Cover, Annual Rainfall, Jan–Feb Rainfall, Mar–May Rainfall, and Jun–Sep Rainfall. The system uses the XGBoost classification algorithm and is integrated with Flask for web deployment.

1.2 Purpose

The purpose of this project is to develop an intelligent early flood prediction system to assist disaster management authorities in taking preventive measures and reducing loss of life and property.

2. IDEATION PHASE

2.1 Problem Statement

Floods cause major environmental and economic damage. Traditional forecasting methods are often inaccurate and slow. A machine learning-based system is required to improve prediction accuracy.

2.2 Empathy Map Canvas

Users: Disaster Management Authorities
Needs: Early flood warnings
Pain Points: Property damage and delayed alerts
Gains: Timely evacuation and reduced losses

2.3 Brainstorming

Multiple ML algorithms such as Decision Tree, Random Forest, KNN, and XGBoost were evaluated. XGBoost was selected based on superior performance metrics.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

User inputs rainfall data → Backend processes → Model predicts → Result displayed.

3.2 Solution Requirement

Functional Requirements:
- Accept rainfall input
- Predict flood occurrence
- Display results

Non-Functional Requirements:
- Accuracy above 90%
- Fast response time
- User-friendly interface

3.3 Data Flow Diagram

User Input → Flask Backend → Data Preprocessing → XGBoost Model → Output Display

3.4 Technology Stack

Python, Flask, XGBoost, Scikit-learn, Pandas, NumPy, HTML, CSS, Joblib

4. PROJECT DESIGN

4.1 Problem Solution Fit

Machine learning identifies rainfall patterns and improves prediction accuracy.

4.2 Proposed Solution

Dataset Collection → Preprocessing → Model Training → Evaluation → Web Integration → Deployment

4.3 Solution Architecture

Dataset → XGBoost Training → Model Save (.save) → Flask Backend → UI → Prediction

5. PROJECT PLANNING & SCHEDULING

Week 1: Dataset Collection
Week 2: Model Training
Week 3: Evaluation
Week 4: Integration
Week 5: Testing and Documentation

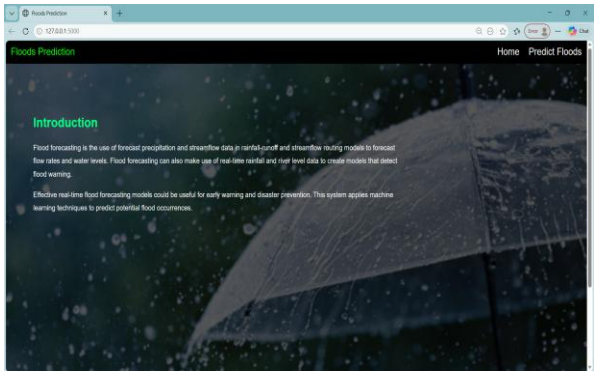
6. FUNCTIONAL AND PERFORMANCE TESTING

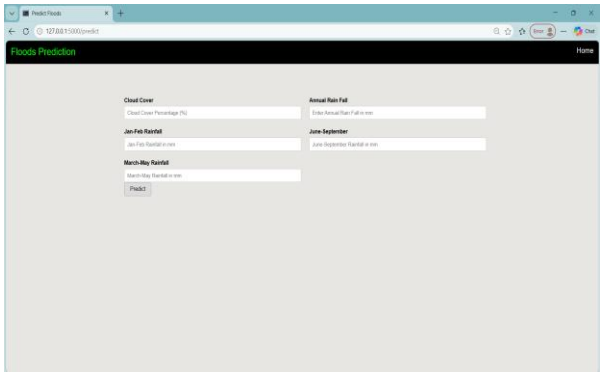
6.1 Performance Testing

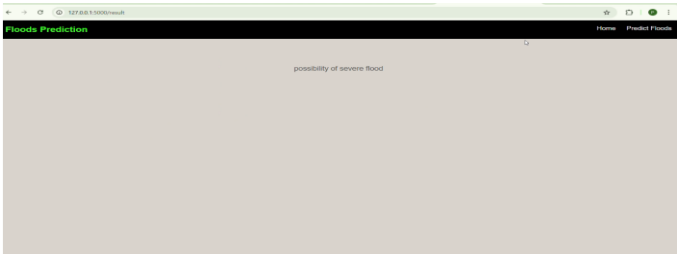
Training Accuracy: 100%
Validation Accuracy: 96.55%
Precision: 100%
Recall: 66%

7. RESULTS

The application successfully predicts flood and no-flood conditions.







8. ADVANTAGES & DISADVANTAGES

Advantages:
- High accuracy
- Fast response
- User-friendly interface

Disadvantages:
- Small dataset
- No real-time weather integration

9. CONCLUSION

The Flood Prediction System demonstrates the effective use of machine learning in disaster management applications.

10. FUTURE SCOPE

- Real-time API integration
- Larger dataset
- Cloud deployment
- SMS alerts

11. APPENDIX

GitHub Repository: https://github.com/vangmayee03/raising-waters/tree/main